MILITARY SPECIFICATION

PROPELLANT, PRIMING FLUID, ALCM ENGINE, GRADE PF-1

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of high density hydrocarbon fuel.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-I-27686 Inhibitor, Icing, Fuel System

MIL-P-87107 Propellant, High Density Synthetic Hydrocarbon Type, Grades JP-9 and JP-10

STANDARDS

Federal

FED-STD-791 Lubricants, Liquid Fuels, and Related Products; Methods of Testing

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-290 Packaging of Petroleum and Related Products

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENESS, Wright-Patterson AFB, OH 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

American Society for Testing and Materials Standards

ASTM D 93 Test Methods for Flash Point by Pensky-Martens Closed Tester

ASTM D 156 Test Method for Saybolt Color of Petroleum Products (Saybolt Chromometer Method)

ASTM D 240 Test Method for Heat of Combustion of Liquid Hydrocarbon Fuels by Bomb Calorimeter

ASTM D 270 Method of Sampling Petroleum and Petroleum Products

ASTM D 381 Test Method for Existent Gum in Fuels by Jet Evaporation

ASTM D 445 Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (and the Calculation of Dynamic Viscosity)

ASTM D 1298 Test Method for Density, Relative Density (Specific Gravity), or API Gravity of Crude Petroleum and Liquid Petroleum Products by Hydrometer Method

ASTM D 1655 Specification for Aviation Turbine Fuels

ASTM D 2276 Test Methods for Particulate Contaminant in Aviation Turbine Fuels

ASTM D 2382 Test Method for Heat of Combustion of Hydrocarbon Fuels by Bomb Calorimeter (High-Precision Method)

ASTM D 2386 Test Method for Freezing Point of Aviation Fuels

ASTM D 3241 Test Method for Thermal Oxidation Stability of Aviation Turbine Fuels (JFTOT Procedure)

ASTM D 3243 Test Methods for Flash Point of Aviation Turbine Fuels by Setaflash Closed Tester

ASTM E 29 Recommended Practice for Indicating Which Places of Figures are to be Considered Significant in Specified Limiting Values

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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Department of Transportation

CFR Title 49, Parts 100-199 — Department of Transportation Rules and Regulations of the Transportation of Explosives and Dangerous Articles.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. Except as otherwise specified herein, the priming fluid shall consist completely of a mixture of exo-tetrahydrodi(cyclopentadiene) and methylcyclohexane.

3.1.1 Reclaimed materials. There is no exclusion to the use of recovered materials. Reclaimed materials shall be required to the maximum extent possible.

3.2 Chemical and physical requirements. The chemical and physical requirements of the priming fluid shall conform to those listed in table I.

3.2.1 Additives. The additives listed shall be used in combination in amounts not to exceed those specified. The type and amount of each additive used shall be reported.

3.2.2 Antioxidants. The following active inhibitors shall be blended separately or in combination into the priming fluid in total concentration of 90 parts per million (minimum) to 110 parts per million (maximum), by weight, not including weight of solvent, in order to prevent the formation of polymeric oxidation products:

a. 2,6-di-tert-butyl-4-methylphenol

b. 6-tert-butyl-2,4-dimethylphenol

c. 2,6-di-tert-butylphenol

3.2.3 Fuel system icing inhibitor. The fuel system icing inhibitor shall conform to MIL-I-27686.

3.3 Performance. The priming fluid shall satisfy the performance requirements as specified in the tests of section 4, herein.

3.4 Workmanship. The priming fluid shall be visually free from undissolved water, sediment or suspended matter and shall be clean and bright at the ambient temperature or at 21°C (70°F), whichever is higher.

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TABLE I. Chemical and physical requirements and test methods.

Property	Priming Fluid		ASTM Standards
	Min	Max	
Color, Saybolt	+25		D 156
Chemical composition, wt. percent			
methylcyclohexane	10.0	12.0	1/
exo-tetrahydrodi (cyclopentadiene)	86.5	90.0	
other hydrocarbons		1.5	
Flash point, °C (°F)	16 (60)	27 (80)	D 93, D 3243
Specific gravity, (60/60F)	0.900	0.930	D 1298
Freezing point, °C (°F)		-54 (-65)	D 2386
Viscosity, centistokes at -54°C		40	D 445
Net heat of combustion MJ/kg (Btu/lb)	42.1 (18,100)		D 240, D 2382
Thermal stability			D 3241 2/
change in pressure drop, mm Hg		10	
heater tube deposit visual rating		code 2	
Existent gum, mg/100 ml		5	D 381
Particulate matter, mg/liter		1.0	D 2276
Fuel system icing inhibitor, volume percent	0.10	0.15	3/

1/ Test shall be performed using Appendix A of MIL-P-87107.

2/ See 4.5.3 for D 3241 test conditions and test limits.

3/ Test shall be performed using method 5327 of FED-STD-791.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections

a. Material inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

4.3 Material inspection. Inspection shall be performed in accordance with method 9601 of FED-STD-791.

4.4 Quality conformance inspection. For acceptance purposes, individual lots shall be examined as specified herein and subject to tests to determine compliance with all requirements in section 3.

4.4.1 Inspection lot

4.4.1.1 Bulk lot. A bulk lot shall consist of an indefinite quantity of a homogeneous mixture of material offered for acceptance in a single isolated container.

4.4.1.2 Packaged lot. A packaged lot shall consist of an indefinite number of 55-gallon drums or smaller unit packages of identical size and shape offered for acceptance and filled from the isolated tank containing a homogeneous mixture of material.

4.4.2 Sampling

4.4.2.1 Sampling for verification of product quality. Each bulk or packaged lot of material shall be sampled for verification of product quality in accordance with ASTM D 270, except where individual test procedures contain specific sampling instructions.

4.4.2.2 Sampling for examination of filled containers for delivery. A random sample of filled containers shall be selected from each lot in accordance with MIL-STD-105 at inspection level II and acceptable quality level (AQL) of 2.5 percent defective. The samples shall be examined in accordance with 4.5.1.3.

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4.5 Inspection methods

4.5.1 Examination

4.5.1.1 Examination of product. Samples selected in accordance with 4.4.2.1 shall be visually examined for compliance with 3.4.

4.5.1.2 Examination of empty containers. Prior to filling, each empty unit container shall be visually inspected for cleanliness and suitability.

4.5.1.3 Examination of filled containers. Samples taken as specified in 4.4.2.2 shall be examined for conformance to MIL-STD-105 with regard to fill, closure, sealing, leakage, packaging, packing, and marking. Any container having one or more defects under the required fill shall be rejected. If the number of defective or underfilled containers exceeds the acceptance number for appropriate plan of MIL-STD-105, the lot represented by the sample shall be rejected.

4.5.2 Chemical and physical tests. Tests to determine conformance to chemical and physical requirements shall be conducted in accordance with table I. The priming fluid shall pass all tests listed in table I. No additional testing shall be required. Requirements contained herein are not subject to corrections for test tolerances. If multiple determinations are made, results falling within any specified repeatability and reproducibility tolerances may be averaged. For rounding off of significant figures, ASTM E 29 shall apply to all tests required by this specification.

4.5.3 Thermal stability test. The thermal stability test shall be conducted using ASTM D 3241. The heater tube shall be rated visually (see Appendix).

4.5.3.1 Test conditions

a. Heater tube temperature at maximum point: 260°C (500°F)

b. System pressure: 3.45 megaPascal (500 pounds per square inch gage)

c. Flow rate: 3.0 milliliter/minute

d. Test duration: 150 minutes

4.5.3.2 Test results. The sample is acceptable if all the following criteria are met:

a. The maximum visual rating of the heater tube deposits is less than a code 3 (Appendix, 30.6).

b. The visual rating of the heater tube shows neither peacock type deposit (code P) nor abnormal type deposits (code A) (Appendix 30.6.3.1 and 30.6.3.2).

c. The maximum differential pressure across the test filter does not exceed 10 millimeters of mercury.

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4.5.3.3 Reported data. The following data shall be reported:

a. Differential pressure in millimeters of mercury at 150 minutes or time to differential pressure of 10 millimeters of mercury, whichever comes first.

b. Heater tube deposit code rating at the end of the test.

4.6 Test report. Test data required by 4.5 shall be reported in accordance with ASTM D 1655, using the standard ASTM form entitled "Inspection Data on Aviation Turbine Fuels" or AFTO Form 476 (see 6.2.2).

5. PACKAGING

5.1 Packaging, packing, and marking. Packaging, packing, and marking shall be in accordance with MIL-STD-290. All containers shall be marked with the actual flash point in degrees F of the priming fluid contained therein.

5.2 Transportation. The transportation of the priming fluid shall be in accordance with CFR title 49, Parts 100-199.

6. NOTES

6.1 Intended use. The priming fluid covered by this specification is intended for use in the ALCM F107-WR-101 engine.

6.2 Ordering data

6.2.1 Procurement documents. Procurement documents shall specify:

a. Title, number, and date of this specification

b. Quantity required and size containers desired

c. Level of packaging and packing required (see 5.1)

d. When blocking and bracing is specified (see 5.2)

6.2.2 Contract data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n)(2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraph:

Paragraph No.	Data requirement	Applicable DID No.
4.6	Quality conformance test reports (fuels)	DI-T-3733

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(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Precaution of mixing inhibitors. To prevent any possible reaction between the concentrated forms of the different inhibitors (see 3.2.1) the contractor is cautioned not to commingle inhibitors prior to their addition to the priming fluid.

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APPENDIX

HEATER TUBE DEPOSIT RATING

10. SCOPE

10.1 Scope. This appendix provides instructions for conducting and reading the heater tube deposit. This appendix is a mandatory part of the specification.

20. REFERENCED DOCUMENTS. This section is not applicable to this appendix.

30. VISUAL TEST METHOD

30.1 Snap the upper end of the heater tube into the clamp of the adapter for the heater tube.

30.2 Push the heater tube against the stop of the adapter tube.

30.3 Slide the adapter with the heater tube over the guide rod into the tuberator equipped with a magnifying glass assembly.

30.4 Insert the ASTM color standard into the tuberator.

30.5 Rotate the adapter and position the heater tube so that the side with the maximum deposit is visible.

30.6 Within 30 minutes after completion of the test, visually examine the heater tube in a tuberator. The entire portion of the test section between the bottom shoulder and the top shoulder of the heater tube test section shall be carefully examined using a magnifying glass in conjunction with the tuberator for any signs of discoloration, scratches, or other visually identified defects. When an area of the tube corresponds visually to an ASTM color standard, that color standard code number shall be recorded. If the area being rated has a color between two adjacent color standards, it shall be rated as the lighter (that is lower number) color standard. (NOTE: It is important that all light bulbs in the tuberator are functioning as a change in light intensity can shift the rating significantly.) (NOTE: The person rating the tube should have normal ability to distinguish between colors; i.e., he should not be color blind.)

30.6.1 In rating the heater tube, the darkest deposits govern and the code number representative of the darkest section, rather than the average deposit, shall be reported.

30.6.2 If a spot or streak is found on the heater tube, it shall be carefully examined under various lighting conditions using a magnifying glass to determine if it is a deposit, a scratch, or tube defect (note that the tube defects should have been found during the pretest inspection of the tube). If the spot or streak is determined to be a scratch or tube defect, it shall be disregarded. If the spot or streak is a deposit, it shall be rated against the ASTM color standards, if larger in area than about 0.004 square inch (0.025 sq cm); i.e.,

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approximately 1/16 inch x 1/16 inch (1.5 mm x 1.5 mm) square or an equivalent area. However, a streak deposit shall be ignored if less than 1/32 inch (0.8 mm) wide, regardless of length. Note that the tube section is 1/8 inch (about 3 mm) in diameter; thus a 1/16 inch (1.5 mm) wide spot is half the diameter of the tube test section and 1/32 inch (0.8 mm) wide streak is one fourth the diameter of the tube test section.

30.6.3 If the heater tube has deposits which do not match the Color Standards, the following criteria shall be used:

30.6.3.1 If the deposit has peacock (rainbow) colors, rate this as code P (P for peacock). If some portion of the deposit does match the Color Standard, it shall be rated.

30.6.3.2 Deposits having abnormal colors (for example, blue or gray) shall have rating of code A (A for abnormal color) assigned.

30.6.3.3 When reporting the overall tube rating, record the rating of the maximum deposit which matches the Color Standards plus P or A if the tube contains deposits which do not match the Color Standards. If the tube contains only P or A deposits, just report the appropriate letter(s); do not try to assign a numerical rating to a P or A deposit. Examples of how the rating procedure is to be used are given below:

Example 1: The darkest deposits on the heater tube match color Standard 3. Also present are peacock colors. Thus, the overall tube rating to be reported is 3P.

Example 2: The heater tube has maximum deposits falling between Color Standards 2 and 3 and has no peacock or abnormal colors. The total tube rating is 2.

Example 3: The heater tube matches Color Standard 1 except for an abnormal deposit which does not match the ASTM Color Standards. The overall tube rating to be reported is 1A.

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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